Sleep Quality After Night Shifts Versus Days Off

A 22-day self-tracking study using R and data visualization

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Research question

Is my sleep quality score different after night shifts than on days off?

Result

After night shifts

74.7

mean score, 9 nights

On days off

83.4

mean score, 11 nights

8.7 points lower after night shifts, but not statistically significant (p = 0.10).

Key findings

- Sleep after night shifts scored lower: the median was 80, against 87 on days off.
- The gap is sizeable (d = −0.78) but uncertain. The 95% interval runs from 19.3 points lower to 1.9 points higher, so the result is inconclusive, not proof of no difference.
- Every sleep after a night shift lasted 5 to 7 hours, and shorter sleeps tended to score lower.

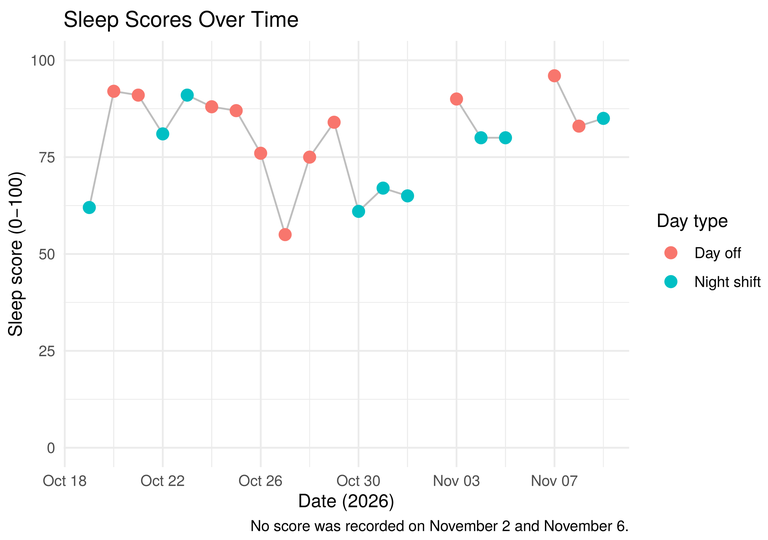


Figure 1. Every recorded score in date order. Scores swing from night to night, and three night shifts in a row (Oct 30 to Nov 1) scored 61 to 67. Gaps mark the two days with no score.

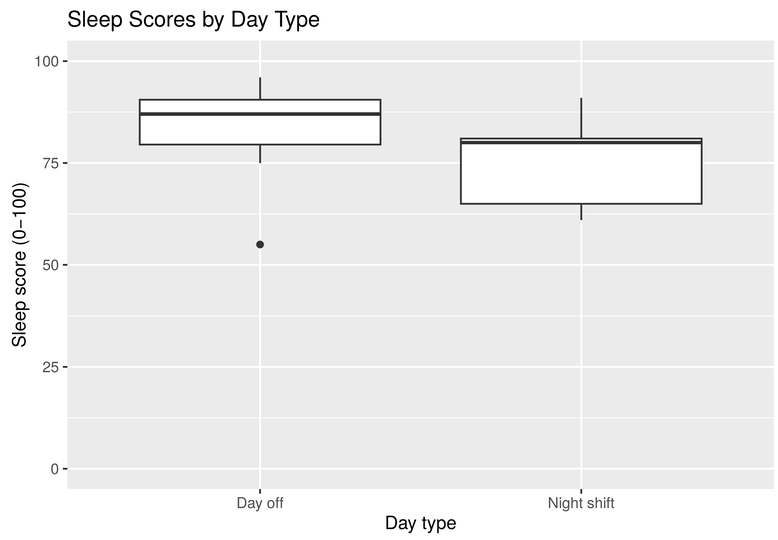


Figure 2. Scores by day type. The median is 87 on days off and 80 after night shifts, but the boxes overlap, and the single lowest score (55) came on a day off.

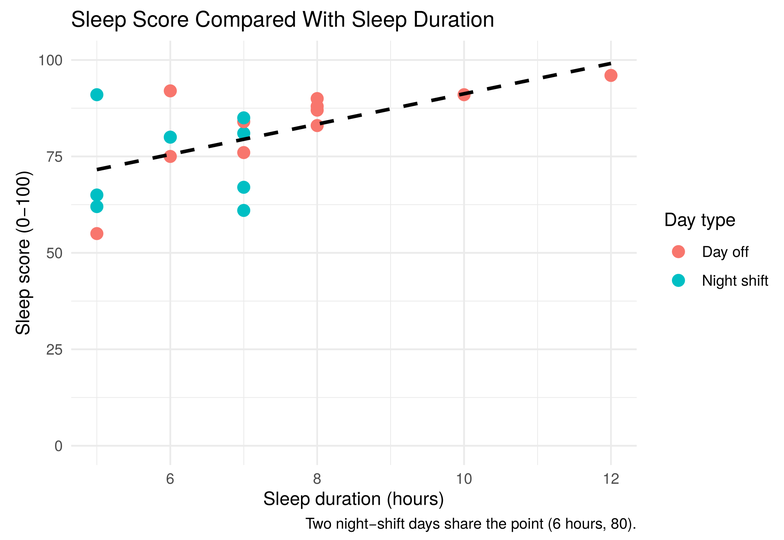


Figure 3. Score against hours slept, with a fitted line. Longer sleeps tended to score higher. Sleeps after night shifts lasted 5 to 7 hours; days off ranged from 5 to 12.

How I collected the data

- 22 consecutive days: October 19 to November 9, 2026.
- One row per day: date, day type (night shift or day off), sleep score (0–100) and hours slept.
- Scores recorded from my Fitbit.
- A night-shift row is the main sleep after that shift.
- November 2 and 6 have no score. I left them blank instead of estimating, which leaves 20 nights.
- Cleaned and graphed in R with readxl, dplyr and ggplot2.

Scores analyzed	Nights	Mean	Lowest	Highest
Night shift	9	74.7	61	91
Day off	11	83.4	55	96

Statistical test

H_0: the mean score is the same after night shifts and on days off.
H_1: the two means differ.

Welch two-sample t-test, two-sided, α = 0.05. It compares the mean of a numeric score between two separate groups of nights and does not assume equal spread.

Difference (night − off)	95% confidence interval	p-value
−8.70	**−19.29 to 1.89**	**0.101**

t statistic	Degrees of freedom	Effect size (d)
−1.73	**17.45**	**−0.78**

Decision: fail to reject H_0. There is not enough evidence that the mean scores differ.

Limitations

- One person and only 20 nights, so the test has little power and the result does not generalize.
- Back-to-back nights are not independent, which the t-test assumes.
- Sleep after night shifts was also shorter, so day type and sleep length cannot be separated.

Conclusion

My sleep scored about 9 points lower after night shifts, but three weeks of data cannot rule out chance. Shorter sleep after a shift may explain part of the gap. More weeks of tracking would give a firmer answer.